RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 8A: Graphing Techniques

1 PJC Prelim 9758/2018/01/Q1

It is given that $\mathrm{f}(x) = 5x^3 + 7x^2 - kx + 2$, where k is a constant.

(i) If the graph of $y = \mathrm{f}(x)$ is strictly increasing, find the range of values of k. [3]

(ii) If $k = 0$, state the range of values of x where the graph of $y = \mathrm{f}(x)$ is concave downward. [1]

2 NJC Promo 9758/2020/Q4

A curve has equation $y + \ln y = \mathrm{e}^x$.

(i) Show that $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(y^2 + y\right) = y^2\mathrm{e}^x + \left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2$. [2]

(ii) Explain why the curve is concave upwards for all real values of x. [2]

3 Show that $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(4x + \sin x - 8\sin\dfrac{x}{2}\right) = 2\cos^2\dfrac{x}{2} - 4\cos\dfrac{x}{2} + 3$. (Hint: $\cos 2\theta = 2\cos^2\theta - 1$)

Hence, by completing the square, show that $4x + \sin x - 8\sin\dfrac{x}{2}$ increases as x increases for all real values of x. Deduce that $4x + \sin x - 8\sin\dfrac{x}{2} > 0$ for $x > 0$. **[5]**

4 RJC JC1 CT 9740/2006/01/Q8

Sketch a continuous curve $y = \mathrm{f}(x)$ with the following properties, labelling the coordinates where possible.

x	y	Behaviour
$x < 2$		Strictly decreasing, Concave upwards
2	1	Stationary point
$2 < x < 4$		Strictly increasing, Concave upwards
4	4	Point of inflexion
$4 < x < 6$		Strictly increasing, Concave downwards
6	7	Stationary point
$x > 6$		Strictly decreasing, Concave downwards

[3]

Rational Functions

5 JPJC Promo 9758/2020/Q4

The curve C has equation $y = \dfrac{2x+a}{x+b}$, for non-zero constants a and b. The graph of C passes through the point with coordinates $(0,1)$. The line $x = 6$ is an asymptote to the graph of C.

(i) Determine the values of a and b. [2]

(ii) Sketch the graph of C, clearly stating the equations of the asymptotes and the coordinates of the axial intercepts. [3]

(iii) By adding a suitable graph to your sketch in **(ii)**, deduce the number of real roots for $2x^3 + ax^2 + x + b = 0$. [3]

6 RVHS JC1 Promo 9758/2019/Q8

The curve C has equation $y = \dfrac{x^2+ax+6}{x-2}$, where a is a constant. It is given that the line $y = x+5$ is an asymptote of C.

(i) Show that $a = 3$. [2]

(ii) Prove algebraically that y cannot lie between -1 and 15. [2]

(iii) Sketch the curve C, stating clearly the coordinates of any points of intersection with the axes, the coordinates of any stationary points and the equations of asymptotes. [3]

(iv) Hence, solve the inequality $\dfrac{x^2+3x+6}{x-2} \geq -x-3$. [2]

7 VJC JC1 Promo 9758/2019/Q10

A curve C has equation $y = \dfrac{\alpha x^2+x+1}{x+2}$, where α is a real, non-zero constant.

Show that if C has 2 stationary points, then $\alpha < 0$ or $\alpha > k$, where k is a constant to be determined. [4]

Sketch the curve C for $\alpha = -1$, giving the equations of asymptotes, the coordinates of stationary points and points of intersection with the axes. [4]

By considering C and an appropriate line, find the range of values of m such that the equation

$$-x^2 + x + 1 = mx^2 + (2m+3)x + 6$$

has exactly 1 positive root and 1 negative root. [3]

8 SRJC Prelims 9740/2009/01/Q7

The curve C has equation $y = \dfrac{x^2 + b}{x + a}$, where $a > 0$ and $b > 0$.

(i) State the coordinates of the intersection(s) of the curve C with the axes in terms of a and b. **[1]**

(ii) Find the equation(s) of the asymptote(s). **[2]**

(iii) Draw a sketch of the curve C, labeling the equation(s) of its asymptote(s) and coordinates of any intersection with the axes. **[2]**

(iv) Hence find the range of values of k, where k is a positive constant, for which the equation $x^2 + b = (x + a)(kx - a)$ has no real root. **[2]**

9 NYJC Promo 9740/2006/01/Q11

The curve C has equation given by $y = \dfrac{ax^2 + bx + c}{x + d}$. It has a vertical asymptote with equation $x = -2$ and its oblique asymptote cuts the x-axis at $x = -2$.

(i) State the value of d and show that $b = 4a$. **[3]**

(ii) Find the condition for a and c such that the gradient of C is always positive. **[3]**

(iii) If C passes through the origin, find the value of c. **[1]**

(iv) Given that $a = 2$ and using the values of b, c and d found in (i) and (iii), sketch the graph of C. **[3]**

(v) Find the equation of the normal line to C at the origin, and determine the coordinates of the point where this normal line cuts C again. **[4]**

Conic Sections

10 VJC JC1 Promo 9758/2019/Q4

The curve C has equation $\dfrac{(y-2)^2}{9} - \dfrac{(x-3)^2}{4} = 1$.

(i) Sketch C, giving the equations of its asymptotes and the coordinates of any turning points.

[4]

The curve D has equation $12y^2 - 48y + 48 + ax^2 - 6ax - 3a = 0$, where a is a positive constant.

(ii) Find the set of values of a for which C and D do not intersect. [2]

11 TMJC Promo 9758/2020/Q4

The curve C has equation $y = \dfrac{2}{x^2 - k^2}$ where $k \geq 2$. It is given that C has only one stationary point.

(i) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. Hence find the coordinates of the stationary point in terms of k. [4]

(ii) Sketch the graph of C, stating the equations of any asymptotes, coordinates of any points of intersection with the axes and the coordinates of the stationary point. [3]

(iii) By adding a suitable graph on the same diagram in part **(ii)**, find the range of values of h, in terms of k, where h is a positive real constant and $h \neq k$ such that $(hx)^2 + \left(\dfrac{2k}{x^2 - k^2}\right)^2 = (kh)^2$ has at least one real root. [3]

12 VJC Prelim 9740/2013/02/Q2(modified)

The curve C has equation $y = \dfrac{ax^2 + bx + c}{x - 1}$, where a, b and c are non-zero constants.

(a) It is given that C passes through the point $\left(3, \dfrac{23}{2}\right)$ and has a minimum point at $(2, 10)$.

(i) Find the values of a, b and c. **[3]**

(ii) Sketch C, giving the coordinates of any turning points, points of intersection with the axes and the equations of any asymptotes. **[3]**

(iii) Find the range of values of m and k, such that the curve with equation $\dfrac{x^2}{k^2} + \dfrac{y^2}{m^2} = 1$ does not intersect C. **[2]**

(b) In the case when $a = -1$ and $b = -1$, find the set of values of c, where $c \neq 2$, such that C has no stationary point. **[3]**

Parametric Equations

13 Sketch, on separate diagrams, the curves defined by the following pair of parametric equations:

(a) $x = a\cos^3\theta, \quad y = a\sin^3\theta$ for $0 \le \theta \le \pi$ where a is a positive constant.

(b) $x = t^2 - 4, \quad y = 3t^4 + 8t^3$ for $t \in \mathbb{R}$;

(c) $x = 1 + \frac{1}{t}, \quad y = t - \frac{1}{t^2}$ for $t \in \mathbb{R}, t \neq 0$

Find the equation of the curves in cartesian form. **[10]**

14 DHS Promo 9758/2018/Q8

A curve C has parametric equations

$$x = \frac{k}{t}, \quad y = t^2 + t,$$

where $t \neq 0$ and k is a positive real constant.

(i) Explain why the lines $x = 0$ and $y = 0$ are asymptotes to C. [2]

(ii) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of k and t. Given that C has exactly one minimum point, show that the coordinates of the minimum point is $(-2k, -\frac{1}{4})$. [4]

(iii) Sketch C. [3]

15 NYJC Promo/9758/2015/Q2

A curve is defined by the parametric equations

$$x = \sin t - \cos t, \quad y = \ln\left(\frac{t}{\pi}\right), \text{ where } 0 < t \le \frac{\pi}{2}.$$

(i) Show that the gradient of the tangent to the curve at the point P, where the curve crosses the y-axis, is $\frac{2\sqrt{2}}{\pi}$. [3]

(ii) What can be said about the curve as $t \to 0$? [1]

(iii) Sketch the graph of the curve, showing clearly its endpoints if any. [2]

(iv) Hence, find the range of values of x for which the curve is strictly increasing. [1]

16 NJC Promo 9740/2006/01/Q3b

A curve is defined parametrically by the equations

$$x = \frac{t}{1+t}\ ,\ y = \frac{t^2}{1+t},\ t \in \mathbb{R}, t \neq -1\,.$$

(i) Find the Cartesian equation of the curve. **[2]**

(ii) Sketch the curve. Label your graph clearly, indicating any asymptote(s) and stationary point(s). **[3]**

17 TJC Promo 9740/2006/01/Q10

A curve C is defined by the parametric equations $x = \sqrt{t},\ y = t - \frac{1}{\sqrt{t}}$, where $t > 0$.

(i) Find the coordinates of the x-intercept of C. **[2]**

(ii) By considering the behaviour of C when $t \to 0$, find the linear asymptote of C. **[1]**

(iii) Sketch C. **[1]**

18 NYJC JC1CT/9758/2020/Q1

A curve C is defined by the parametric equations $x = \cos\theta + \mathrm{e},\ y = \sin\frac{\theta}{2}$ for $0 < \theta \leq \frac{\pi}{2}$.

(i) Sketch C, giving the exact coordinates of the end-points. [2]

(ii) Find the cartesian equation of C. [2]

(iii) On the same diagram as drawn in **(i)**, sketch the graph $\sqrt{2}y + 2\mathrm{e}x - y^2 - x^2 = \mathrm{e}^2 - \frac{1}{2}$. Hence find the coordinates of the point(s) where the two graphs intersect. [4]